



Entomotherapy

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Abstract

Entomotherapy is the use of insects and their products for medicinal purposes, practiced across cultures since ancient times. Insects like bees, ants, fruit flies, maggots, silkworms, beetles, grasshoppers, termites, crickets, mealybugs, stink bugs, cockroaches and head lice treat infections, cancer, inflammation, respiratory and digestive disorders and wounds. Their bioactive compounds like peptides, alkaloids, enzymes and hormones provide immunological, antibacterial, analgesic and regenerative benefits, making them valuable for modern drug development

Keywords: Entomotherapy, medicinal insects, traditional medicine.

Introduction :

India, with its rich insect biodiversity, has many species used medicinally, though worldwide over 1,000 insect species are known for therapeutic use (Mozui *et al.*, 2021). Entomotherapy, from Greek words meaning “insect treatment,” is the preventive or therapeutic use of insects and their products, practiced since ancient times in various cultures. Insects provide important medicinal compounds with immunological, analgesic, antibacterial, diuretic, anesthetic and anti-rheumatic properties, contributing to both traditional and modern medicine.

Why to study insects as medicine?

Insects and their products have long been used in traditional medicine and provide immunological, analgesic, antibacterial, diuretic, anesthetic and anti-rheumatic benefits, making them valuable for modern drug development. Their by-products remain important in folk medicine and hold potential for discovering new drugs, with no resistance issues reported so far (Chowdary *et al.*, 2022).

Different insects used as medicine (Gavin and Davranoglou 2023)

A. Honey bee

Honey, worth over 1.7 billion dollars annually, has been collected for over 6,000 years and contains sugars, water, vitamins, minerals, amino acids, and enzymes. It has long been used globally including in Egypt, Greece, Rome, traditional Chinese medicine and Ayurveda to treat wounds, digestive and respiratory problems.

B. Ant (Solenopsin)

Ant venom, especially Solenopsin A from fire ants (*Solenopsis invicta*), shows anti-cancer potential by blocking the PI3K/Akt pathway, making it promising for treating cancers like melanoma and ovarian cancer. *Polyrachis lamellidens* is used in Chinese medicine for its powerful analgesic and anti-inflammatory effects.

C. Fruit fly

Fruit flies (*Drosophila melanogaster*) have about 50 types of olfactory receptors that detect specific chemical signals. Since cancer cells release unique metabolic markers, fruit flies can sense these changes, making them useful in detecting cancer. Their natural sensory system could inspire advanced biosensors that outperform current electronic detection methods.

D. Maggots

Studies of cellular and humoral immunity of the blow fly, *Calliphora vicina* maggot revealed three groups of pharmacologically active substances that are used in medicine: alloferons, allostatis and antimicrobial peptides.

E. Silkworm

Silkworms (*Bombyx mori*) are used in traditional medicine for detoxification, infections and compounds from pupae that enhance nitric oxide to treat impotence. Their larvae serve as bioreactors for antioxidant powders, while silk-based materials like curcumin nanoparticles and sericin gels aid cancer therapy and wound healing.

F. Beetle products

Blister beetles (Meloidae) produce cantharidin, used in Chinese medicine for warts and cancer, which inhibits leukemic cell growth and targets leukemia stem cells. Cantharidin derivatives also show anticancer activity against prostate, colon, cervical, gallbladder, and other cancer cell lines.

G. Grasshopper

Grasshoppers have been used in traditional medicine as a diuretic for kidney diseases and to reduce swelling. They are also used to treat intestinal disorders, headaches (by applying grasshopper ash with salt on incisions), and as a tea for asthma and hepatitis. These practices remain part of folk healing in regions like Oaxaca, Mexico.

H. Termite

Termites are widely used in traditional medicine, with 45 species recorded for food and therapeutic purposes. Species like *Microcerotermes exiguous* are used to treat asthma, bronchitis, flu and whooping cough. *Nasutitermes macrocephalus* is used for respiratory problems such as coughs, sore throat, sinusitis and tonsillitis.

I. Cricket

Crickets are traditionally used as a diuretic for dropsy, for treating infected wounds with mole cricket gut contents, and as food to support mental development and maternal care. These practices highlight their medicinal and nutritional importance in folk medicine.

J. Mealy bugs

Mealybugs have been used in traditional medicine as ointments or boiled masses to treat leprosy, skin diseases, burns, swelling, pain, and digestive issues. *Dactylopius coccus* is also valued as a natural dye and used for skin treatments, tooth cleaning, and dental care.

K. Stink bug

Bug oil has been traditionally applied for scrofula, tuberculosis, and ailments of the kidney, liver, and stomach. Live bugs were used as pain relievers, for toothache, rheumatism, gastrointestinal issues, goitre (due to iodine content), and even as an aphrodisiac.

L. Cockroach

Cockroach brain and nerve tissues contain compounds that can kill over 90% of harmful bacteria like *E. coli* and MRSA, offering potential alternatives to antibiotics. Traditionally, cockroaches have also been used to treat ailments such as earache, ulcers, dropsy, whooping cough, epilepsy, and boils.

M. Head lice

In Spanish folk medicine, head lice were traditionally ingested-often hidden in foods like chocolate-to treat jaundice. This practice, passed down orally, has survived in tradition though modern research no longer documents its use.

References

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